

9.21 Solution: From Eq. (9.137), the time-averaged angular momentum density is given by

$$\mathbf{m} = \frac{1}{2c^2} \text{Re} [\mathbf{r} \times (\mathbf{E} \times \mathbf{H}^*)],$$

whose z -component is

$$m_z = \hat{\mathbf{z}} \cdot \mathbf{m} = \frac{1}{2c^2} \text{Re} [\hat{\mathbf{z}} \cdot \{\mathbf{r} \times (\mathbf{E} \times \mathbf{H}^*)\}] = \frac{1}{2c^2} \text{Re} [(\hat{\mathbf{z}} \times \mathbf{r}) \cdot (\mathbf{E} \times \mathbf{H}^*)].$$

It is straightforward to verify that $\hat{\mathbf{z}} \times \mathbf{r} = r\hat{\boldsymbol{\phi}}$, and

$$(\mathbf{E} \times \mathbf{H}^*)_{\phi} = -\frac{k}{Z_0\beta} E_z E_{\phi}^*.$$

Therefore,

$$m_z = \frac{1}{2c^2} \text{Re} \left[r \left(-\frac{k}{Z_0\beta} \right) E_z \left(-\frac{m\beta}{\gamma^2} \right) \frac{E_z^*}{r} \right] = \frac{km}{2Z_0c^2\gamma^2} J_m(\gamma r)^2.$$

The time averaged energy density is given by Eq. (9.134),

$$\begin{aligned} u &= \frac{\varepsilon_0}{4} (|\mathbf{E}|^2 + Z_0^2 |\mathbf{H}|^2) \\ &= \frac{\varepsilon_0}{4} \left(|E_z|^2 + \frac{m^2\beta^2}{\gamma^4} \frac{|E_z|^2}{r^2} + \frac{\beta^2}{\gamma^4} \left| \frac{\partial E_z}{\partial r} \right|^2 + \frac{k^2}{\beta^2} |E_{\phi}|^2 + \frac{k^2}{\beta^2} |E_r|^2 \right) \\ &= \frac{\varepsilon_0}{4} \left(J_m(\gamma r)^2 + \frac{m^2\beta^2}{\gamma^4} \frac{J_m(\gamma r)^2}{r^2} + \frac{\beta^2}{\gamma^4} \left(\frac{dJ_m(\gamma r)}{dr} \right)^2 + \frac{k^2 m^2}{\gamma^4} \frac{J_m(\gamma r)^2}{r^2} + \frac{k^2}{\gamma^4} \left(\frac{dJ_m(\gamma r)}{dr} \right)^2 \right) \\ &= \frac{\varepsilon_0}{4} \left(J_m(\gamma r)^2 + \frac{m^2(k^2 + \beta^2)}{\gamma^4} \frac{J_m(\gamma r)^2}{r^2} + \frac{k^2 + \beta^2}{\gamma^4} \left(\frac{dJ_m(\gamma r)}{dr} \right)^2 \right). \end{aligned}$$

The energy per unit length in the z -direction is then

$$\begin{aligned} \langle u \rangle &= \int u da \\ &= \frac{\varepsilon_0}{4} 2\pi \int_0^R r \left(J_m(\gamma r)^2 + \frac{m^2(k^2 + \beta^2)}{\gamma^4} \frac{J_m(\gamma r)^2}{r^2} + \frac{k^2 + \beta^2}{\gamma^4} \left(\frac{dJ_m(\gamma r)}{dr} \right)^2 \right) dr \\ &= \frac{\pi\varepsilon_0}{2\gamma^2} \int_0^{\gamma R} x \left(J_m(x)^2 + \frac{m^2(k^2 + \beta^2)}{\gamma^2} \frac{J_m(x)^2}{x^2} + \frac{k^2 + \beta^2}{\gamma^2} \left(\frac{dJ_m(x)}{dx} \right)^2 \right) dx. \end{aligned}$$

This can be simplified by the Bessel's equation,

$$\frac{d^2 J_m(x)}{dx^2} + \frac{1}{x} \frac{dJ_m(x)}{dx} + \left(1 - \frac{m^2}{x^2} \right) J_m(x) = 0,$$

or

$$\frac{d}{dx} \left[x \frac{dJ_m(x)}{dx} \right] = -x \left(1 - \frac{m^2}{x^2} \right) J_m(x).$$

The last term in the integral can therefore be expressed as

$$\int_0^{\gamma R} x \left(\frac{dJ_m(x)}{dx} \right)^2 dx = \int_0^{\gamma R} x \frac{dJ_m(x)}{dx} dJ_m(x)$$

$$\begin{aligned}
&= x J_m(x) \frac{dJ_m(x)}{dx} \Big|_0^{\gamma R} - \int_0^{\gamma R} J_m(x) \frac{d}{dx} \left[x \frac{dJ_m(x)}{dx} \right] dx \\
&= \int_0^{\gamma R} x \left(1 - \frac{m^2}{x^2} \right) J_m(x)^2 dx.
\end{aligned}$$

Then,

$$\begin{aligned}
\langle u \rangle &= \frac{\pi \varepsilon_0}{2\gamma^2} \int_0^{\gamma R} x \left(J_m(x)^2 + \frac{m^2(k^2 + \beta^2)}{\gamma^2} \frac{J_m(x)^2}{x^2} + \frac{k^2 + \beta^2}{\gamma^2} \left(1 - \frac{m^2}{x^2} \right) J_m(x)^2 \right) dx \\
&= \frac{\pi \varepsilon_0}{2\gamma^2} \left(1 + \frac{k^2 + \beta^2}{\gamma^2} \right) \int_0^{\gamma R} x J_m(x)^2 dx \\
&= \frac{\pi \varepsilon_0 k^2}{\gamma^4} \int_0^{\gamma R} x J_m(x)^2 dx,
\end{aligned}$$

where we have used the relation $k^2 = \beta^2 + \gamma^2$.

On the other hand,

$$\langle m_z \rangle = \int m_z da = \frac{\pi k m}{Z_0 c^2 \gamma^2} \int_0^R r J_m(\gamma r)^2 dr = \frac{\pi k m}{Z_0 c^2 \gamma^4} \int_0^{\gamma R} x J_m(x)^2 dx.$$

Thus,

$$\frac{\langle m_z \rangle}{\langle u \rangle} = \frac{\pi k m / Z_0 c^2}{\pi \varepsilon_0 k^2} = \frac{m}{Z_0 \varepsilon_0 c^2 k}.$$

Notice that $Z_0 \varepsilon_0 = \sqrt{\mu_0 \varepsilon_0} = 1/c$, the final result becomes

$$\frac{\langle m_z \rangle}{\langle u \rangle} = \frac{m}{ck} = \frac{m \hbar}{\hbar \omega}.$$

This is reasonable, as the electromagnetic field has an angular momentum of $m \hbar$ and energy of $\hbar \omega$.